

Premium Study Notes and Cheat Sheets

Common Preparation / DSA & Placement Prep

Study Hub Premium Placement Pack

This is original study material for preparation. It is not a copied official or proprietary company paper.

Premium Study System

- Track: Placement Common Preparation.
- Pack type: NOTES / Study Material.
- Target outcome: selection-ready confidence for Graduate Engineer / Analyst.
- Core preparation areas: aptitude, reasoning, verbal, DSA, CS fundamentals, resume, project interview, mock tests, communication, HR stories.
- Use this pack like a mini-book: concept, table, example, practice, analysis, revision.
- Every topic should produce proof: solved questions, mock score, mistake note, interview answer, or resume evidence.

Output Quality Rules

- Read the concept in your own words, then solve without looking at notes.
- Convert every important idea into a comparison table, checklist, or one-line rule.
- Mark important traps as Common Mistake before the mock, not after losing marks.
- Keep a PYQ-style log: question pattern, topic, time taken, mistake reason, retest date.
- Use simple English; add Hindi/Hinglish meaning in brackets when it helps recall.

Priority Matrix

Area	Premium Standard	Proof Before Moving On
Aptitude	Formula, shortcut, accuracy, time control	25 mixed questions with 70%+ accuracy
Programming Logic	Loops, arrays, strings, SQL basics	Trace output and write simple code cleanly
Project	Problem, architecture, your contribution, tradeoff	2-minute story plus deep-dive answers
Communication	Crisp intro, structured HR stories, calm tone	5 recorded answers reviewed once

Exam Tip

- Service assessments reward speed with accuracy. Attempt high-confidence sections first and protect time for easy marks.

Common Mistake

- Spending too long on one aptitude question reduces total score. Set a strict skip rule and revisit only after easy wins.

Must Revise

- Aptitude formulas: percentage, ratio, speed-time-distance, time-work, probability, permutation-combination.
- DSA patterns: two pointers, sliding window, hashing, prefix sum, binary search, recursion, graph BFS/DFS, DP states.
- CS core: process/thread, deadlock, indexing, joins, normalization, TCP/UDP, HTTP, OOP pillars.
- Interview habits: clarify, think aloud, dry run, test edge cases, explain complexity.

Premium Study Material Structure

- Concept note: one paragraph in your own words.

- Formula/pattern note: exact rule, when to use, and when not to use.
- Worked example: one solved sample with each step visible.
- Interview angle: how the topic can be asked verbally.
- Revision hook: one mnemonic or one-line trigger.

Memory Tricks

- APTI: Assumption, Pattern, Time limit, Insight shortcut.
- STAR for HR: Situation, Task, Action, Result.
- PACT for projects: Problem, Architecture, Contribution, Tradeoff.

One-Liners For Revision

- Accuracy grows from mistake analysis, not from collecting random PDFs.
- A topic is ready only when you can teach it without opening notes.
- A resume skill is safe only when you can answer two follow-up questions.

Execution Dashboard

Metric	Target	Review Frequency
Aptitude Sets	5 timed sets per week	every Sunday
CS Core Notes	4 topics per week	twice a week
Mock Tests	1-2 per week	same day analysis
Interview Practice	5 answers per week	record and review

Weekly Review Ritual

- Pick the top 5 mistakes, rewrite the correct rule, and solve one similar question.
- Remove one weak resume line or prepare one stronger project answer.
- Decide the next week from evidence, not mood.

Final Self Audit

Score	Meaning	Action
0-40	reading stage	finish basics and easy questions
41-70	practice stage	increase timed mocks and mistake review
71-85	interview-ready	polish project, HR, and weak topics
86-100	selection-ready	maintain revision and avoid overloading new material

Source Note

- This is original Study Hub premium material for learning and practice. It is not copied from official or proprietary company papers.